

Adossi Fred William

Software Engineering student · Aspiring AI Research Engineer

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I investigate meaningful problems and turn research into responsible, reliable AI systems, with a particular interest in healthcare and agriculture across African contexts.

My goal is to become an AI Research Engineer, combining the curiosity and rigour of a researcher with the practical discipline of an engineer to investigate important problems and translate validated ideas into responsible, reliable systems.

EDUCATION

BSc (Hons) Software Engineering

African Leadership University, Kigali, Rwanda

June 2024 to an expected finish in July 2027

June 2024 to an expected finish in July 2027, with a CGPA of 4.14 out of 5.00 and all 380 attempted credits earned, as recorded on my August 2026 transcript.

Web Infrastructure 98.80%

Machine Learning Pipeline 94.26%

Frontend Web Development 93.80%

Enterprise Innovation Program 92.50%

Foundations Project 89.82%

Enterprise Web Development 86.10%

Mathematics for Machine Learning 84.42%

Mobile Application Development 83.80%

VERIFIED ACADEMIC RECORD

4.14 / 5.00

Current CGPA

380 / 380

Attempted credits earned

View unofficial ALU transcript →

Transcript printed 27 August 2026.

RESEARCH DIRECTION

From meaningful questions to systems that work

I hope to establish BRTI as an African research and technology organisation where researchers, AI and software engineers, healthcare professionals, agricultural experts, entrepreneurs, and communities investigate real challenges together and transform rigorous research into practical solutions.

Responsible AI

Healthcare

Agriculture

Research capacity

SELECTED RESEARCH AND ENGINEERING PROJECTS

LEADERSHIP AND COMMUNITY

- As the founder and facilitator of the AI Study Lab, I have watched it grow from twelve members to more than thirty across Chad, Cameroon, and Rwanda, and I reshaped it into an on-demand support model once the way people were actually using it told me that was what they needed.
- Through community mentorship between Chad and Rwanda, I have guided more than ten students into places at ALU, the University of Rwanda, and CMU Africa, while advising over twenty others along the way.
- I am also a member of the Claude Builder Club at ALU, where the focus is on building artificial intelligence that serves a genuine social purpose.

AWARDS AND INTERESTS

- I took second place at the Claude Builder Club Hackathon at ALU in 2026 for an AI-powered social impact solution, which I am still developing well beyond the competition itself.
- Long before any of this, I ran a self-funded natural juice business in N'Djamena, where I handled everything from sourcing and production through to pricing and selling.
- Away from engineering I keep a steady routine of basketball and daily strength training, and my wider interests reach into artificial intelligence, financial inclusion, public speaking, and drawing.

TECHNICAL PREPARATION

Malaria Diagnosis with CNN and Transfer Learning

[ML](#) · [Healthcare](#)

Investigates how compact and transfer-learning CNNs compare on malaria cell images, with my baseline model reaching 0.9647 F1 on the held-out test set.

African Credit Scoring System

[ML](#) · [Financial Inclusion](#)

Explores alternative signals for financial inclusion by modelling bank-account access across 23,524 East African survey responses—with its limits stated clearly.

HealthBridge Africa

[Full-stack](#) · [Healthcare](#)

Explores inclusive telemedicine through mobile, web, and an offline USSD pathway, with separate AI support designed for clinicians and patients.

PeerForge

[Full-stack](#) · [Developer Tools](#)

A collaboration environment where computer science students can find projects, share work, and form teams instead of learning in isolation.

Rice Leaf Disease Classifier

[ML](#) · [Agriculture](#)

Helps investigate four rice-leaf diseases from a photograph through a deployed, retrainable pipeline tested under simulated concurrent demand.

Productivity Tracking and Analytics App

[Mobile](#) · [Productivity](#)

Turns personal work logs into private, on-device insight, while exposing model quality and separating learned predictions from rule-based recommendations.

EXPERIENCE

Machine Learning Intern

2025

[CodeAlpha](#)

- I designed and deployed an African Credit Scoring System and made a point of building explainability into it with SHAP and LIME, then containerised the whole solution with Docker so that it was genuinely ready for production.
- I also built a Speech Emotion Recognition pipeline that combined CNN and LSTM architectures on the RAVDESS dataset, and I relied on audio augmentation together with a speaker-independent split so that no information could leak between training and evaluation.

Field Research Consultant and Web Developer

2025

[Afrifarmers Market](#)

- I carried out structured field research with around thirty cooperative members in Rwamagana District, and I turned what I learned from them directly into the requirements for the product we would go on to build.
- After delivering a consultancy report on the findings, I was trusted with a follow-on contract, which gave me the chance to build a web application for managing operations and tracking transactions as they happen.

Data Analysis Intern

2024

[MTN Mobile Money \(MoMo\)](#)

- Having only just arrived in Rwanda, I secured the role by going to MTN's head office in person, finding my way to the right contact, and following up with an email and a formal interview.
- Once I was inside, I parsed and processed Mobile Money transaction records stored in XML using Python, and I built an interactive dashboard that presented the insights clearly to colleagues who did not come from a technical background.

Founder and Lead Developer

2025 to Present

[Independent Web Development Practice](#)

- Alongside full-time study, I run my own client-facing development practice, and over the past year it has grown to more than fifteen paid projects spanning web applications, business websites, portfolios, and design work.
- I take responsibility for the entire lifecycle of each engagement, which starts with gathering requirements and scoping the work and continues through design, delivery, and handover, and when a contract is large enough I bring together and coordinate a team of four to see it through.
- One of my clients was based in Nigeria, and I delivered their portfolio within four days, while a separate engagement saw me build a multi-language institute website at a four-figure contract value.

Shipped to real users

Python · JavaScript ES6+ · TypeScript · Flutter · Dart · React · Next.js · Node.js · FastAPI · Firebase · Firestore · Riverpod · Google Maps SDK · Docker · Streamlit · HTML5 · CSS3 · Tailwind · SQL · MySQL · SQLite · Git · Linux · Nginx · XML processing

Built and evaluated in projects

TensorFlow · Keras · scikit-learn · CNN · LSTM · VGG16 · ResNet50 · MobileNetV2 · XGBoost · Random Forest · Logistic Regression · K-Means · PCA · SHAP · LIME · pandas · NumPy · librosa · NestJS · Prisma · Socket.IO · Clerk · Turborepo · Flask · PostgreSQL · REST API design · JWT auth · GitHub Actions · Postman · R

SELECTED CERTIFICATIONS

Neural Networks and Deep Learning

DeepLearning.AI and Coursera · May 2026

Linear Algebra for ML and Data Science

DeepLearning.AI and Coursera · Jun 2026

Flutter and Dart for iOS, Android and Mobile Apps

IBM and Coursera · Mar 2026

Data Analysis with R Programming

Google and Coursera · Jan 2026

Claude 101

Anthropic · 2026

AI for Everyone

DeepLearning.AI and Coursera · Jul 2026

REFERENCES AND VALIDATION

Academic documentation, project source code, demonstrations, and professional references are available for scholarship review.

Agricultural Field Worker and Operations Trainee

2019 to 2023

Family Farming Business

- For four years I worked across several farm sites, where my responsibilities ranged from preparing the land and operating tractors to harvesting and maintaining relationships with the farmers around us.
- During that time I was trained in farm management and commercial negotiation, and the problems I saw at close range are the very same ones that continue to shape the agricultural technology I build today.